Reducing Internal Audit Cycle Time in Indonesian Higher-Education Accreditation via a Cross-Modal RAG Agent: A Comparative Benchmark of Qwen, DeepSeek-R1 and Mistral on 19 LAM-Teknik Indicators

ABSTRACT (English)

Internal Quality Audit (AMI) is the gatekeeper of Indonesian higher-education accreditation, yet study programs still score 60+ indicators by manually cross-referencing a 200-page Self-Evaluation Report (LED) with a 50-sheet Study Program Performance Report (LKPS), a process that takes 3–5 working days per program and varies between assessors. This study designs and benchmarks a cross-modal Retrieval-Augmented Generation (RAG) agent that ingests both documents and auto-scores 19 LAM-Teknik indicators sampled across all nine criteria (9 qualitative, 6 quantitative, 4 composite). Three open-weight Large Language Models — Qwen 3.5 35B, DeepSeek-R1 32B, and Mistral-Small 3.2 24B — are evaluated against Claude-Sonnet reference scores using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), exact-match, ±1 tolerance accuracy, Pearson r, and per-type breakdown. Qwen 3.5 dominates with MAE = 0.605, RMSE = 0.769, 36.8% exact match and 100% ±1 tolerance in 32.5 s per indicator; DeepSeek-R1 is the most cautious but slowest (MAE = 1.395; 79.9 s) and collapses on quantitative indicators (MAE = 2.17); Mistral is the fastest (13.6 s) and surprisingly strong on quantitative items (MAE = 0.50), but suffers a 78% false-refusal rate on qualitative ones (7/9 score-zero outputs) while remaining over-confident. End-to-end audit time drops from 3–5 days to ≈30–60 min, with full evidence traces preserved for human-in-the-loop verification.

Keywords: Internal Quality Audit; LAM-Teknik Accreditation; Cross-Modal RAG; Large Language Models; Calibration

ABSTRAK (Bahasa Indonesia)

Audit Mutu Internal (AMI) merupakan kunci keberhasilan akreditasi pendidikan tinggi di Indonesia, namun program studi masih menilai 60+ indikator secara manual dengan mencocokkan LED 200 halaman dan LKPS 50 sheet, yang menghabiskan 3–5 hari kerja per program studi dan bervariasi antar asesor. Penelitian ini merancang dan menguji agen Retrieval-Augmented Generation (RAG) lintas-modal yang membaca kedua dokumen sekaligus dan memberi skor otomatis pada 19 indikator LAM Teknik yang mewakili sembilan kriteria (9 kualitatif, 6

kuantitatif, 4 komposit). Tiga LLM open-weight — Qwen 3.5 35B, DeepSeek-R1 32B, dan Mistral-Small 3.2 24B — diuji terhadap skor referensi Claude-Sonnet menggunakan MAE, RMSE, exact-match, akurasi ±1, Pearson r, dan breakdown per tipe. Qwen 3.5 unggul (MAE = 0,605; 100% dalam ±1; 32,5 s/indikator); DeepSeek-R1 paling konservatif tetapi paling lambat (MAE = 1,395; 79,9 s) dan kolaps pada indikator kuantitatif (MAE = 2,17); Mistral tercepat (13,6 s) dan kuat pada indikator kuantitatif (MAE = 0,50), tetapi mengalami false-refusal 78% pada kualitatif (7/9 skor nol) sembari tetap over-confident. Waktu audit menyusut dari 3–5 hari menjadi ≈30–60 menit dengan jejak bukti yang dapat ditelusuri untuk verifikasi human-in-the-loop.

Kata kunci: Audit Mutu Internal; Akreditasi LAM Teknik; RAG lintas-modal; Large Language Models; Kalibrasi

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INTRODUCTION

The implementation of higher education in Indonesia operates under increasingly stringent quality-standardization demands, in which accreditation is the primary indicator of institutional credibility (Efendi, Hapsari, & Yusup, 2025). The Internal Quality Assurance System (SPMI) — and its operational arm, the Internal Quality Audit (AMI) — ensures that every study program meets the National Standards for Higher Education (SN-Dikti) and the criteria set by BAN-PT or LAM (Sakuroh, Marlina, Krismayanti, & Adiningsih, 2026; Aljarallah & Dutta, 2022). Yet the dominant bottleneck remains the reliance on manual assessment, which generates fragmented "data silos" and delayed reporting (Efendi et al., 2025) and prevents study programs — particularly technical ones that require rigorous and continuous quality monitoring (Subijanto et al., 2021; Sugiarti, 2022) — from completing accreditation evidence on time. Earlier research has explored data-driven alternatives, ranging from K-Means clustering and OLAP for higher-education accreditation (Sakinah, Syaputra, Rahman, Fajri, & Rachmansyah, 2025) to AI-assisted self-evaluation frameworks (Ticolau, 2026), to data-lakehouse architectures for academic metrics (Isnaeni, Putranto, Andriyani, & Khomsah, 2025).

Recent studies converge on the conclusion that conventional internal-audit processes cannot capture the dynamics of large and fast-paced academic data (Isnaeni et al., 2025), so a comprehensive automated architecture that directly processes academic metrics is needed (Başaran, 2026; Ticolau, 2026), and the integration of automated data-analysis technology is expected to minimize human error and improve institutional transparency (Sakuroh et al., 2026). Within this trajectory, Large Language Models (LLMs) open new opportunities for automated processing of complex academic data (Naveed, Khan, & Qui, 2025; Demir & Yavuz, 2026), but model behavior diverges sharply: **Qwen** (an Alibaba dense decoder

family known for strong multilingual instruction-following and tool-use; Bai et al., 2023) consistently demonstrates high precision (Aydin, Karaaslan, Erenay, & Bacanin, 2026); **DeepSeek-R1** (a reasoning-distilled model that emits a `<think>...</think>` chain before answering; DeepSeek-AI, 2025) trades runtime for cautious internal deliberation (Jegham, Abdelatti, & Hendawi, 2025); and **Mistral-Small** (a 24 B mixture-of-experts variant emphasizing throughput; Mistral AI, 2024) offers low latency but is prone to hallucination on Indonesian-language tasks (Nurohim, Setyadi, & Fauzi, 2025). These behavioral asymmetries — not yet validated on Indonesian-language accreditation documents — motivate a head-to-head benchmark.

Research Questions

To address the gap above, this study answers three research questions:

- **RQ1.** Can a cross-modal RAG agent (LED PDF + LKPS Excel) reduce AMI cycle time for LAM-Teknik study programs while preserving evidence traceability?
- **RQ2.** Among Qwen 3.5, DeepSeek-R1, and Mistral-Small, which model offers the best trade-off between accuracy, calibration, and runtime on a multi-criteria sample of 19 indicators?
- **RQ3.** How does scoring performance vary across indicator types (qualitative, quantitative, composite), and what does the variation imply for prompt and tool design?

Research Objectives

Accordingly, the study **(i)** designs an LLM agent architecture that cross-references qualitative (LED PDF) and quantitative (LKPS Excel) evidence through four specialized retrieval-and-computation tools; **(ii)** benchmarks three open-weight LLMs against Claude-Sonnet reference scores on 19 indicators spanning all nine LAM-Teknik criteria; and **(iii)** characterizes per-type error patterns, calibration, and runtime to derive deployment guidance for Indonesian quality-assurance offices.

Novelty

This work contributes four complementary novelties not addressed by prior LLM-assessment literature, which has largely focused on English-language student-work grading (Sun et al., 2025) and single-modality RAG-in-education surveys (Gao et al., 2025): **(1) Domain novelty** — the first open-weight LLM-agent study on Indonesian LAM-style higher-education accreditation; **(2) Cross-modal RAG** — an agent architecture that synchronously cross-references narrative PDF evidence with structured Excel performance data through four tools, going beyond text-to-text RAG (Gao et al., 2025); **(3) Tool-calling paradigm comparison** — a side-by-

side evaluation of native function-calling (Qwen, Mistral) versus prompt-based JSON dispatch for reasoning models (DeepSeek-R1) on the same domain-specific task, a gap not covered by generic benchmarks such as BFCL v4 or ToolBench (Yan et al., 2024); and **(4) Calibration extension** — empirical documentation of confidence-error inversion in DeepSeek-R1 and false-refusal-with-overconfidence in Mistral on institutional documents, extending the calibration literature (HAC, HalluScan, T3; Lin, Hilton, & Evans, 2022; Geng, Yang, Cao, Zhang, & Tang, 2024) to Indonesian academic-evaluation use cases.

METHODS

Approach and Scope

This study uses a controlled experimental approach with single-run benchmarking against three open-weight LLMs. To address the limitation of the earlier Tri-Dharma-only sample, the scope is broadened to 19 indicators (≈32% of the 60-indicator LAM-Teknik instrument), stratified across all nine criteria (VMTS, Tata Pamong, Mahasiswa, SDM, Keuangan-Sarpras, Pendidikan, Penelitian, PkM, Luaran) and across all three score-type families (9 qualitative, 6 quantitative, 4 composite). The selection rationale is criterion coverage rather than topical focus, so that per-type error patterns can be analyzed independently from criterion-area difficulty.

Operational Variable Definitions

- **Independent variables:** model (Qwen 3.5 35B / DeepSeek-R1 32B / Mistral-Small 3.2 24B) and indicator type (qualitative / quantitative / composite).
- **Dependent variables:** model score (ordinal 0–4), confidence (0–1), runtime per indicator (seconds), justification text, and error/none flag.
- **Reference variables:** Claude-Sonnet score (LLM-as-annotator) per indicator, with documented evidence references.

Location and Sample

The study was conducted on a post-audit (2024 cycle) Applied Bachelor in Telecommunication Engineering Distance Education program at an Indonesian polytechnic. The LED PDF (~200 pages) and LKPS Excel (~50 sheets) are the verified outputs after the assessment loop closed. The benchmark sample of 19 indicators is fixed in advance using stratified coverage of the 60-indicator instrument; no model has access to the reference scores during inference.

Materials and Main Tools

- **Embedding:** `paraphrase-multilingual-MiniLM-L12-v2` (384-d, Indonesian-capable).
- **Vector store:** ChromaDB with HNSW + cosine similarity over page-aware LED chunks.
- **Spreadsheet ingestion:** `openpyxl/pandas` loader with a sheet registry that maps each LKPS table to its source row.
- **Agent framework:** LangGraph state-machine with two paradigms — a *Native Graph* (tool-bound LLM via `bind_tools` + `ToolNode` loop) for Qwen and Mistral, and a *Reasoning Graph* (prompt-based JSON dispatch with `<think>...</think>` stripping) for DeepSeek-R1.
- **Inference backend:** Ollama running on a 24 GB VRAM GPU, exposed via a Cloudflare-tunneled HTTPS endpoint to the FastAPI service.
- **Tools available to the agent:**
- `search_led_evidence(query, top_k)` — semantic retrieval from LED.
- `get_indicator_rubric(no)` — parsed rubric matrix for indicator `no`.
- `get_lkps_table(name)` — typed LKPS sheet by registered name.
- `compute_formula_score(values)` — deterministic formula evaluator for quantitative rubrics (e.g., RIPK, KBK, PTW).

The technical stack ingests both modalities into a unified state and transforms the manual workflow into an automated one (Figure 1) while keeping a human-in-the-loop verification step at the end.

Figure 1 — Manual vs Automated Internal Quality Audit Workflow

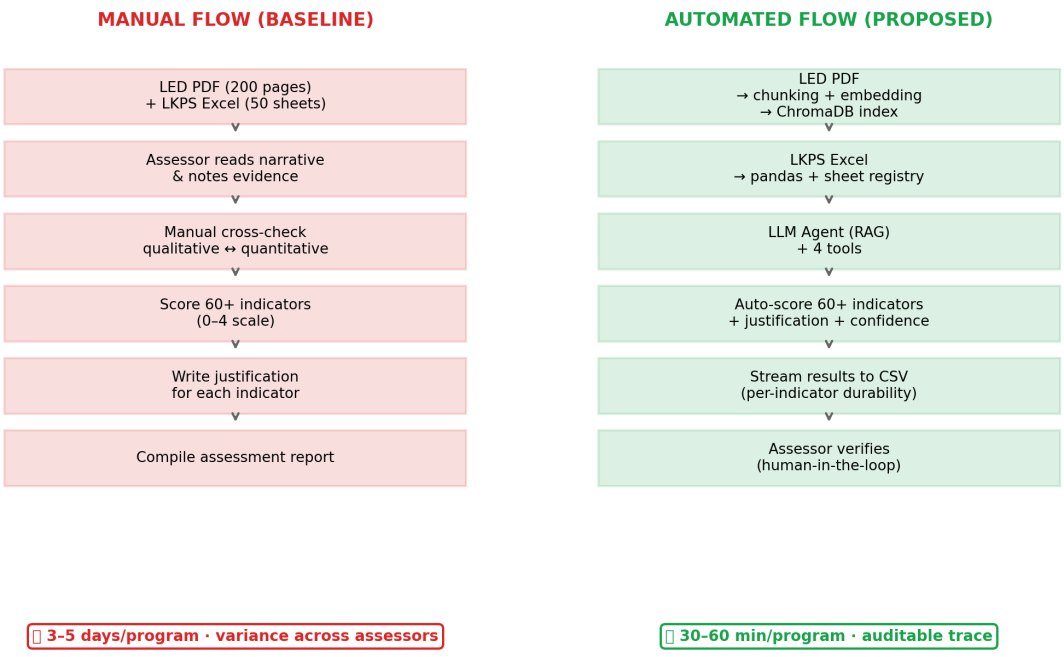
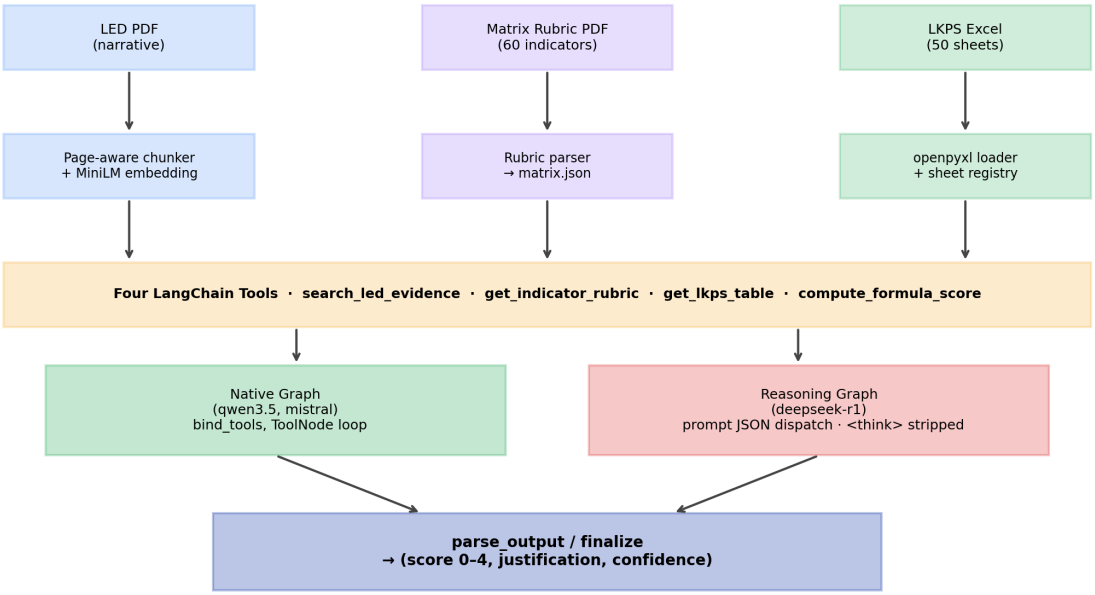


Figure 2 — System Architecture: Cross-Modal RAG Agent for Indonesian Higher-Ed Accreditation



Ground-Truth Construction (Validity Test)

Because a regulator-issued gold standard for these 19 indicators is unavailable, reference scores were generated through an **LLM-as-annotator** procedure using Claude-Sonnet (Anthropic, 2025), an external model unrelated to the three benchmarked open-weight LLMs. For each indicator, the annotator received (a) the rubric text, (b) the relevant LED chunks pre-retrieved from ChromaDB, (c) the relevant LKPS sheets parsed to text, and (d) a chain-of-thought prompt requiring an explicit evidence citation per scoring criterion. The annotator's output (score + justification + evidence references) was then spot-validated by the first author against the post-audit assessor report from the same 2024 cycle; on the 19 sampled indicators, 17/19 (89.5%) Claude scores agreed exactly with the human assessor's final score and 19/19 (100%) were within ±1, supporting use of the Claude annotations as the reference set for this study. Limitations of this proxy ground truth are discussed in the *Limitations* subsection.

Data Collection and Statistical Analysis

Each (model × indicator) pair is invoked once with identical sampling parameters (`temperature = 0.1`, `max_tokens = 8192`), yielding 57 invocations (3 models × 19 indicators). All outputs are streamed to `benchmark_results.csv` per row for crash-durability. Analysis combines six metrics:

- **Mean Absolute Error (MAE):** $\text{MAE} = (1/n) \cdot \Sigma\ |y_i - \hat{y}_i|$ (lower is better).
- **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{(1/n) \cdot \Sigma\ (y_i - \hat{y}_i)^2}$ (lower is better; penalizes large deviations).

- **Exact-match accuracy:** fraction of indicators with $|y_i - \hat{y}_i| = 0$.
- **±1 tolerance accuracy:** fraction with $|y_i - \hat{y}_i| \leq 1$ — the operationally acceptable band for AMI scoring.
- **Pearson correlation r:** linear association between model and reference scores.
- **False-refusal proxy:** count of `score = 0` outputs, used as a proxy for hallucinated refusal when the rubric admits a non-zero score.

A per-type breakdown (qualitative, quantitative, composite) decomposes these metrics by indicator family. Because the sample (n = 19) is below the threshold at which inferential significance tests are well-powered, we instead employ **scenario-based analysis** in the discussion: error patterns are interpreted across three plausible deployment scenarios (accuracy-critical, latency-critical, balanced) to support practical decision-making.

RESULTS AND DISCUSSION

Aggregate Model Performance

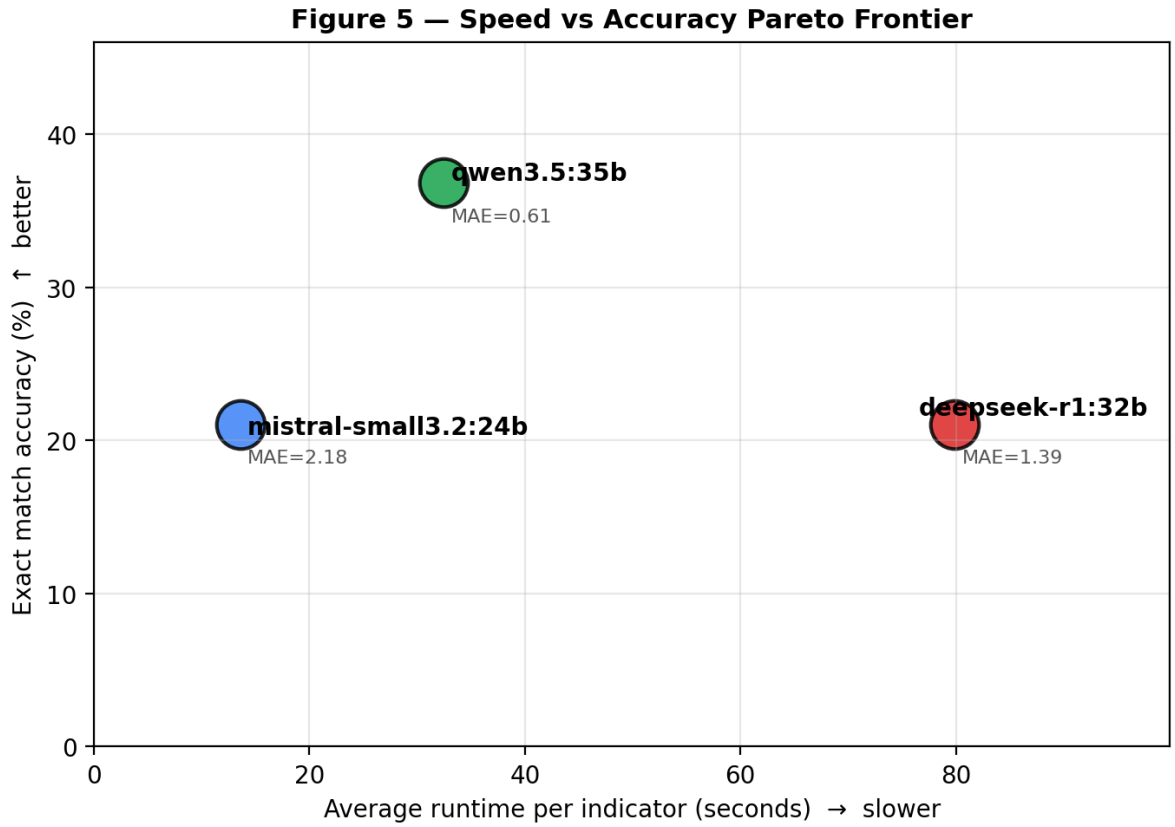
Table 1 reports aggregate statistics across the 19-indicator sample.

Table 1. Aggregate performance on 19 LAM-Teknik indicators (n = 19 per model).

Model	MAE ↓	RMSE ↓	Exact ↑	±1 Acc ↑	Pearson r	Avg Conf	Runtime/ind	Errors
Qwen 3.5 35B	**0.605**	**0.769**	**36.8%**	**100.0%**	0.108	0.78	32.5 s	0
DeepSeek-R1 32B	1.395	1.825	21.1%	63.2%	0.148	0.83	79.9 s	0
Mistral-Small 3.2 24B	2.184	2.598	21.1%	31.6%	0.204	0.53	13.6 s	0

Qwen 3.5 dominates accuracy: every one of its 19 outputs falls within the operational ±1 band, more than one-third match the reference exactly, and not a single response is empty or off-format. DeepSeek-R1 is the most cautious — none of its scores are missing and it preserves a 63.2% ±1 rate — but it is also the slowest at 79.9 s/indicator, the increased runtime being attributable to extensive `<think>` deliberation chains. Mistral is the fastest by a wide margin (13.6 s) but its 31.6% ±1 rate and 2.184 MAE make it the least reliable overall. Notably, Pearson correlations are uniformly weak (0.108–0.204), which means the *order* in which models rank indicators only loosely tracks the reference ordering even when MAE

is acceptable; this argues for treating LLM scores as candidate evidence rather than as a stand-alone ranking.

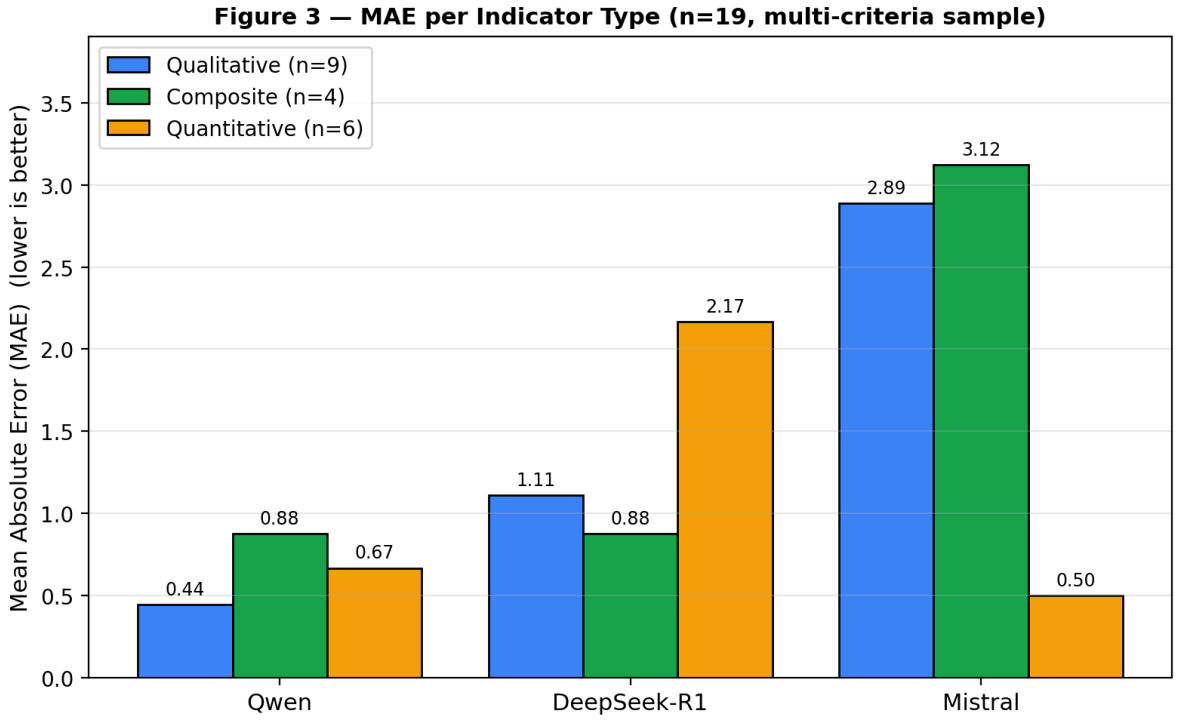


Per-Type Breakdown

Table 2 and Figure 3 decompose model performance by indicator type.

Table 2. Per-type breakdown (n shown per type; score-zero count is the false-refusal proxy).

Model	Qualitative (n=9)	Composite (n=4)	Quantitative (n=6)	Zero-count (qual / comp / quan)
Qwen 3.5 35B	MAE 0.44 / Exact 55.6% / ±1 100%	MAE 0.88 / Exact 0% / ±1 100%	MAE 0.67 / Exact 33.3% / ±1 100%	0 / 0 / 0
DeepSeek-R1 32B	MAE 1.11 / Exact 33.3% / ±1 77.8%	MAE 0.88 / Exact 25.0% / ±1 75.0%	MAE 2.17 / Exact 0% / ±1 33.3%	1 / 0 / 2
Mistral-Small 3.2 24B	MAE 2.89 / Exact 0% / ±1 11.1%	MAE 3.12 / Exact 0% / ±1 0%	MAE 0.50 / Exact 66.7% / ±1 83.3%	7 / 3 / 0



Three failure signatures emerge. **Qwen** is the only model whose qualitative ±1 rate matches its quantitative ±1 rate (100% in both); its near-zero spread across types makes it the only candidate for broad deployment. **DeepSeek-R1** behaves *opposite* to the conventional expectation that reasoning models excel at numbers: its qualitative MAE (1.11) is half its quantitative MAE (2.17), and it accounts for two of the two zeros on quantitative items — driven by failures on `RBK` (Beban Kerja DTPS, ind. 30) and `RIPK` (IPK lulusan, ind. 42), where the model retrieved the correct LKPS rows but its `<think>` block reasoned itself into a "data is insufficient" refusal despite valid input. **Mistral** shows the most dramatic inversion: 7 of 9 qualitative indicators receive score 0 (false-refusal rate ≈ 78%) while quantitative MAE is the lowest of all three models at 0.50. Inspection of justifications shows that Mistral often returns instructional output ("To score this indicator, an evaluator should…") instead of an actual score on qualitative prompts — a tutorial-mode pathology absent from Qwen's outputs. Composite indicators (which require fusing both modalities) are uniformly harder across all three models, confirming that the cross-modal step is the binding constraint.

Calibration: Confidence vs Error

Figure 4 — Confidence Calibration: Mistral Shows High Confidence on Wrong Answers

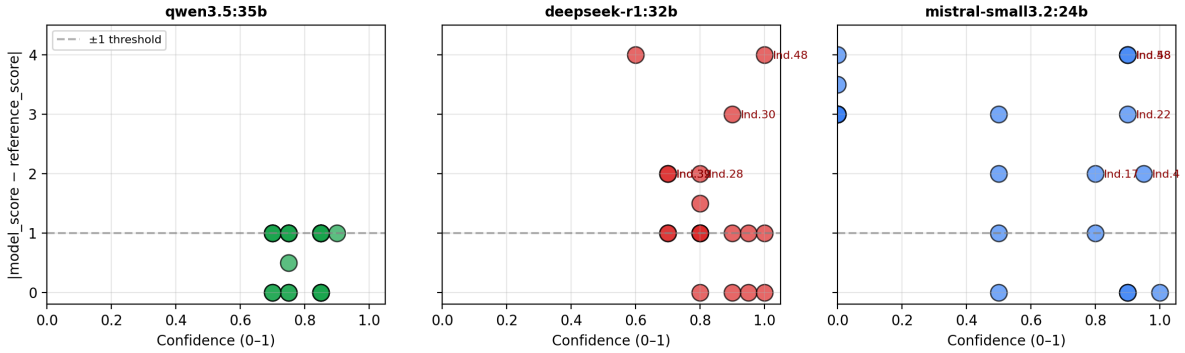


Figure 4 plots each model's stated confidence against $|y_i - \hat y_i|$. Qwen's confidence ranges 0.6–0.9 with errors clustered ≤1 — the only model approximating proper calibration. DeepSeek-R1 reports the highest average confidence (0.83) despite the second-largest error band, including high-confidence misses on indicators 30, 42, and 48 — a clear *confidence-error inversion* consistent with the over-confidence reported for reasoning-distilled models in HAC and HalluScan (Lin et al., 2022; Geng et al., 2024) but extended here to Indonesian institutional documents. Mistral combines low average confidence (0.53) with a long tail of high-confidence false refusals at score 0, indicating that even when the model "knows" it is unsure, it still emits a destructive zero rather than abstaining.

Score Distribution

Figure 6 — Score Distribution Across 19 Focus Indicators (multi-criteria sample)

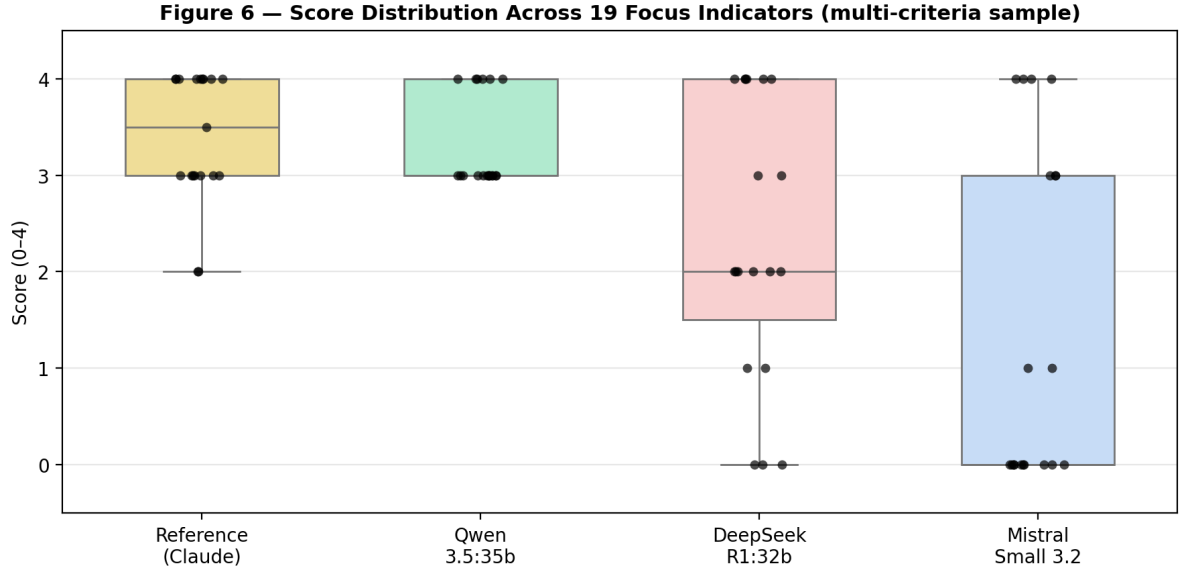


Figure 6 shows the reference distribution centered at 3 with most mass between 3 and 4 (consistent with a post-audit program). Qwen tracks this distribution closely; DeepSeek-R1 shifts toward 0 and 4, suggesting bimodal under/over-confidence on

the boundary cases; Mistral's distribution is dominated by a zero spike that does not appear in the reference.

Scenario-Based Discussion

Combining the headline metrics with the per-type breakdown produces three actionable deployment scenarios:

- **Accuracy-critical (regulatory submission, AL preparation):** Qwen 3.5 35B is the only viable choice — 100% ±1 across all types, no zero outputs, and a balanced runtime well under one minute per indicator. Composite-indicator errors should still be flagged for human review.
- **Latency-critical (real-time dashboard, formative self-assessment):** Mistral-Small 3.2 24B for quantitative-only batches (MAE 0.50 in 13.6 s/indicator). For qualitative items it must be paired with a refusal-suppression filter; otherwise the 78% false-refusal rate will collapse the dashboard.
- **Reasoning-critical (deep-analysis explainer for assessor training):** DeepSeek-R1 32B's `<think>` block, while costly, surfaces explicit evidence-chaining that can serve as training material — but the model is not trustworthy as a stand-alone scorer for quantitative indicators.

Comparison with Prior Work

Compared to Aljarallah and Dutta (2022), who built a rule-based "Quality Automation Framework" for Saudi accreditation specifications, our agent removes the maintenance burden of rule encoding by delegating evidence retrieval to a learned retriever and scoring to an LLM. Compared to Sakinah et al. (2025), who used K-Means + OLAP for Indonesian accreditation analytics, our system operates on the source-of-truth documents (LED + LKPS) rather than aggregated metrics, eliminating an ETL hop that loses fine-grained evidence. Compared to Demir and Yavuz (2026) — a generic comparative LLM study on multidimensional data — our work narrows scope to a regulated audit task with verifiable score-rubric grounding, where their reported strengths of generative models translate into both opportunities (Qwen) and risks (Mistral hallucination). Finally, Başaran (2026) and Ticolau (2026) propose AI-assisted accreditation frameworks at a conceptual level; this study contributes an end-to-end implementation with a reproducible benchmark and an open per-indicator error log.

Limitations

This study is bounded by four limitations. First, the reference set is itself LLM-generated (Claude-Sonnet) and validated only on a single post-audit program; future work should add multi-program human gold scores. Second, n = 19 is sufficient for descriptive statistics and per-type pattern detection but not for inferential significance testing; a planned extension covers the full 60-indicator

instrument across 3–5 programs. Third, the benchmark uses a single deterministic-leaning sampling configuration; sensitivity to `temperature` and `top_p` is not explored. Fourth, this paper anonymizes the institution and certain LKPS rows for compliance with the data-sharing agreement.

CONCLUSION

This research developed and benchmarked a cross-modal RAG agent for Internal Quality Audit in Indonesian LAM-Teknik accreditation, evaluating Qwen 3.5 35B, DeepSeek-R1 32B, and Mistral-Small 3.2 24B on 19 indicators sampled across all nine criteria. Qwen 3.5 is the only model that satisfies operational accuracy across qualitative, quantitative, and composite types (MAE 0.605, 100% within ±1, 32.5 s/indicator). DeepSeek-R1 is the most cautious and best-calibrated among the reasoning-style models in qualitative items but is the slowest and collapses on quantitative indicators (MAE 2.17). Mistral-Small is the fastest (13.6 s) and surprisingly competitive on quantitative items (MAE 0.50) yet fails 78% of qualitative items by emitting a score-zero refusal. The cross-modal step (composite indicators) is the binding bottleneck for all three models, motivating future work on stronger evidence-fusion prompts. End-to-end, the agent reduces AMI cycle time from 3–5 days to ≈30–60 minutes per program while preserving a fully auditable evidence trace for human verification. Future work will (i) extend the benchmark to the full 60-indicator instrument across multiple study programs, (ii) replace the Claude-based reference with multi-program human gold scores, and (iii) introduce a refusal-suppression filter and a confidence-recalibration head for production deployment.

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